

General Description

This dataset contains single-cell tracking data of HeLa cervical cancer cells obtained via long-term time-lapse microscopy using a custom shear-free, diffusive microfluidic platform. The data was used for parameter estimation and testing a cellular automata (CA) model to analyze cell cycle variability and inheritance.

File Structure

The dataset consists of 15 individual CSV files, each representing data from one of the 15 independent microfluidic chambers analyzed in the study (e.g., `1.csv` to `15.csv`).

Data Format and Column Descriptions

Each CSV file contains the following columns:

- **Position X [μm]**: The horizontal (X) coordinate of the cell's center of mass, measured in micrometers.
- **Position Y [μm]**: The vertical (Y) coordinate of the cell's center of mass, measured in micrometers.
- **Time [h]**: The elapsed time from the start of the experiment, recorded in hours.
- **Lineage ID**: A unique hierarchical identifier (pedigree) encoding the cell's division history:
 - **Founder cells** (cells present at the start of the observation) are assigned single-digit IDs (e.g., 1, 2, 3).
 - **Daughter cells** inherit their mother's ID with an appended 1 or 2 upon division (e.g., cell 1 divides into 11 and 12; cell 11 subsequently divides into 111 and 112).
 - The total number of digits in the ID directly corresponds to the generation index of the cell.

Dataset Partitioning

For the purposes of the iterative parameter-testing and model training described in the manuscript, the chambers were divided as follows:

- **Training Set**: Data from chambers 9, 10, and 11.
- **Test Set**: Data from all other chambers (1–8 and 12–15).

Contact Information

For any questions regarding the dataset structure, please contact the corresponding author as indicated in the main manuscript.